Physiotherapy OSCE Preparation

Cardiorespiratory — Cases C & D | Target: 70%+ (Very Good)

Case C: Non-CF Bronchiectasis | Case D: Heart Failure (HFrEF) | 5 Sections per Case

Ms. Sarah Collins

58-year-old Female · Non-CF Bronchiectasis Exacerbation · Admitted with increased cough & sputum over 2 weeks

CASE C

1 Subjective Assessment

CONSENT / OPENING STATEMENT

> *"Hello Ms. Collins, my name is [Name] and I'm one of the physiotherapists. I've been asked to assess your breathing and cough today. I'd like to ask you some questions and then do a brief physical assessment — nothing should be uncomfortable, and you're free to stop at any time. Do I have your permission to proceed?"*

KEY QUESTIONS + CLINICAL REASONING

- **Can you describe your sputum — the colour, thickness, and how much you're producing compared to your usual?** *(Sputum characteristics are the primary physiotherapy driver in bronchiectasis. Green or purulent sputum in increased volume signals an active infective exacerbation and guides ACT intensity, positioning choice, and urgency of medical review re: antibiotics.)*
- **Have you noticed any blood in your sputum at all?** *(Haemoptysis is a red flag in bronchiectasis — it may indicate erosion of bronchial artery vessels, a known complication. This changes my treatment plan completely, as percussion and vigorous ACT are contraindicated.)*
- **How breathless are you compared to your normal level — are you short of breath getting dressed or walking to the bathroom?** *(Establishes functional baseline and degree of deterioration from usual status. Helps me classify exacerbation severity and decide whether assisted ACT or supplemental O_2 monitoring is needed.)*
- **Which airway clearance techniques do you normally use, and what do you feel is stopping them from working as well right now?** *(Understanding her usual self-management routine identifies barriers —*

e.g., increased sputum viscosity, fatigue, dyspnoea — so I can target the specific problem, whether that is technique modification, positioning, or an adjunct device.)

- **Do you have any chest pain, particularly when you breathe in deeply or cough?** *(Pleuritic chest pain suggests pneumonia or pneumothorax — both change my treatment plan and require urgent medical review before physiotherapy proceeds.)*
- **How is the fatigue affecting your ability to do your airway clearance and your usual daily activities?** *(Fatigue during exacerbation is a major self-management barrier. This informs pacing strategies, session length, and whether I need to initially assist with ACT rather than expecting full independent performance.)*

RED FLAGS

Haemoptysis (frank blood in sputum)

Indicates possible erosion of bronchial artery — a known life-threatening complication of bronchiectasis. Volume matters: streaking vs. massive haemoptysis carry different urgency.

Action: Stop ACT immediately. Do NOT perform percussion or vigorous expiratory techniques. Notify medical team immediately.

SpO_2 <92% at rest / Acute severe dyspnoea

Suggests acute respiratory failure — possible from mucus plugging, pneumothorax, or pneumonia superimposed on bronchiectasis. Physiotherapy cannot proceed safely.

Action: Sit patient upright. Apply supplemental O_2 as per prescription. Call medical emergency.

Systemic sepsis (fever >38.5°C, confusion, hypotension, tachycardia >120 bpm)

Severe infective exacerbation requiring immediate medical management. The patient is not safe for active physiotherapy in this state.

Action: Defer physiotherapy. Alert medical team. Document and reassess once medically stabilised.

Sudden severe chest pain with absent unilateral breath sounds

Possible pneumothorax — a recognised complication of severe bronchiectasis. ACT is absolutely contraindicated.

Action: Stop all treatment immediately. Urgent CXR and medical review. Emergency call if haemodynamically compromised.

PROBLEM LIST

PHYSICAL IMPAIRMENTS

- Excess retained sputum in bronchiectatic airways
- Increased sputum viscosity and purulence
- Airflow obstruction secondary to mucus plugging

FUNCTIONAL ACTIVITY LIMITATIONS

- Exertional dyspnoea — reduced exercise tolerance
- Impaired ability to self-manage ACT effectively

- Impaired mucociliary clearance
- Reduced respiratory muscle endurance
- Generalised deconditioning from infection/inflammation

- Fatigue limiting ADL performance
- Disrupted sleep from nocturnal cough
- Reduced participation in community activities

2 Pathophysiology

Bronchiectasis is characterised by **permanent, abnormal dilation and distortion of the bronchi**, resulting from a self-perpetuating cycle of infection, inflammation, and airway wall destruction. In non-CF bronchiectasis, repeated insults (infection, immune dysfunction) damage the structural components of the bronchial wall — the elastic tissue and smooth muscle — leading to irreversible dilation.

The dilated airways are structurally abnormal and **cannot effectively propel mucus** via the mucociliary escalator. Mucus stagnates within these pouches and pools, providing an ideal culture medium for bacterial colonisation — commonly *Haemophilus influenzae*, *Pseudomonas aeruginosa*, or *Staphylococcus aureus*. This triggers a sustained neutrophilic inflammatory response, with neutrophil proteases (elastase, MMP-9) further destroying the airway wall — the classic "vicious cycle of bronchiectasis."

During an exacerbation, increased bacterial load leads to **heightened inflammation, increased sputum production, and changed sputum rheology** — sputum becomes thicker, more viscous, and purulent, making it even harder to clear. Progressive mucus accumulation causes **segmental atelectasis and V/Q mismatch**, reducing oxygenation and increasing the work of breathing.

Physiotherapy targets the mucus retention limb of this cycle. By mobilising and clearing retained secretions through ACT, we aim to break the infectious cycle, reduce airway obstruction, restore ventilation to collapsed segments, and slow the progressive airway damage.

Why Ms. Collins fits: Her 2-week history of increased sputum volume that is thick and hard to clear, worsening exertional breathlessness, and reduced ACT efficacy are the hallmarks of a bronchiectasis exacerbation. The known structural diagnosis explains the substrate; the change in sputum character reflects worsening bacterial load and airway inflammation. V/Q mismatch from mucus plugging explains her increasing exertional dyspnoea.

3 Differential Diagnoses

Differential 1: Community-Acquired Pneumonia (CAP)

Supporting features

- Productive cough with purulent sputum over 2 weeks
- Worsening dyspnoea and systemic fatigue
- CXR may show new consolidation

- Possible systemic features: fever, elevated inflammatory markers

Differential 2: COPD Exacerbation with mucus hypersecretion

Supporting features

- 58-year-old female with obstructive airway disease pattern
- Productive cough, dyspnoea, and reduced exercise tolerance
- Fatigue and reduced ACT efficacy
- Possible co-existing COPD in a patient with chronic airway disease

I favour Bronchiectasis Exacerbation over Community-Acquired Pneumonia because Ms. Collins has a known, established diagnosis of non-CF bronchiectasis with a characteristic pattern of recurrent exacerbations. Her 2-week subacute deterioration with change in sputum character — rather than an acute febrile illness — is more consistent with an exacerbation of her underlying condition than a new pneumonic episode. Pneumonia typically presents more acutely and would not explain reduced ACT efficacy as the primary physiotherapy concern.

I favour Bronchiectasis Exacerbation over COPD Exacerbation because the primary and most prominent feature is sputum volume and quality — the defining physiotherapy problem in bronchiectasis. While COPD can cause mucus hypersecretion, the known structural diagnosis, the nature of the secretion-clearance failure, and the characteristic ACT self-management history all point firmly to a bronchiectasis exacerbation as the primary diagnosis driving this admission.

4 Objective Assessment — Special Tests

Test 1: Sputum Assessment (Colour, Consistency, Volume)

- **What it measures:** Sputum quality and quantity — the primary physiotherapy treatment driver in bronchiectasis. Directs ACT technique selection, session frequency, and flags need for medical review.
- **How to perform:** Ask the patient to expectorate into a clear sputum pot after morning ACT. Observe: colour (clear → mucoid → mucopurulent → purulent green/brown), consistency (watery, viscous, tenacious), and volume (mL/session or 24-hour collection). Compare to patient's stated baseline. Document using the Murray Sputum Score or descriptive grading.
- **Positive finding:** Green/purulent sputum in increased volume (e.g., >30 mL/day above baseline) = active infective exacerbation requiring assisted ACT and medical review regarding antibiotic therapy.
- **Why I select this:** I choose sputum assessment because in bronchiectasis, the sputum characteristics are our primary treatment guide. Understanding what we are trying to clear informs my choice of ACT technique, gravity-assisted drainage position, and session frequency. It also provides a measurable outcome to track treatment response.

Expected Auscultation: Coarse, wet, low-pitched crackles — bilateral or localised to affected lobes — most prominent on early-to-mid inspiration. Importantly, crackles may partially clear after a productive cough or huff (distinguishes from pulmonary oedema crackles, which do not clear). Possible monophonic wheeze if airway inflammation is significant.

RR: 18–24 bpm (mildly elevated) **SpO_2: 92–96% on air** **Accessory muscles: mild use**

Outcome Measure: Sputum diary — daily volume (mL) and colour grade. Track each session through admission and at outpatient follow-up.

Test 2: Peak Expiratory Flow Rate (PEFR) / Bedside Spirometry (FEV_1)

- **What it measures:** Degree of airflow obstruction — quantifies exacerbation severity, establishes baseline, and monitors physiotherapy response over the admission.
- **How to perform:** PEFR — patient seated upright, takes maximal inspiration, forms a tight seal around the mouthpiece, and blows out as fast and hard as possible. Record the best of 3 attempts. Compare to personal best or predicted value (% predicted). If spirometry available: measure FEV_1 and FVC pre and post bronchodilator.
- **Positive finding:** PEFR <80% predicted or significant fall from personal best = meaningful airflow obstruction. FEV_1/FVC <0.70 = obstructive pattern consistent with bronchiectasis with mucus plugging.
- **Why I select this:** I choose PEFR/spirometry because knowing the degree of airflow limitation helps me assess exacerbation severity objectively and informs whether pre-ACT bronchodilator therapy may improve secretion clearance by opening airways. It also gives me a baseline to track improvement, which is important for discharge decisions.

Expected Auscultation: Widespread coarse crackles — may be more prominent post-spirometry due to increased tidal volume dislodging secretions (a useful productive sign). Possible expiratory wheeze. Diminished air entry at bases if significant atelectasis is present.

Assess RR post-test **SpO_2 throughout test** **Stop if desaturation >4%**

Outcome Measure: FEV_1 (% predicted) — repeat at end of admission. Target return toward admission baseline or pre-exacerbation spirometry values if known.

Test 3: 1-Minute Sit-to-Stand Test (1MSTS)

- **What it measures:** Functional exercise capacity and physiological response to exertion — a validated, safe alternative to the 6MWT during acute exacerbation when a corridor walk may be too demanding.
- **How to perform:** Patient sits in a standard chair (approximately 45 cm height), arms crossed over chest. On "Go," patient stands fully erect and returns to sitting as many times as possible for 60 seconds. Count

complete repetitions. Record HR, SpO_2, and Borg dyspnoea and fatigue scores immediately before and within 30 seconds of completion. Stop early if SpO_2 <88%, chest pain, dizziness, or Borg >7/10.

- **Positive / Impaired finding:** <20 repetitions in a 58-year-old female (normative ≈ 25–30) = significantly reduced functional capacity. SpO_2 drop >4% = exercise-induced desaturation, requiring supplemental O_2 during rehabilitation.
- **Why I select this:** I choose the 1MSTS because Ms. Collins is acutely unwell and a full 6MWT is likely unsafe and too demanding at this stage. The 1MSTS requires only a chair, takes 60 seconds, can be stopped instantly if needed, and still gives me valid, reproducible information to baseline functional capacity and guide my exercise prescription for rehabilitation.

Expected Auscultation (post-test): Increase in coarse crackles due to increased tidal volume mobilising peripheral secretions — this is a potentially useful sign, indicating physiotherapy is moving secretions. Monitor for new wheeze or marked reduction in air entry as signs of exercise intolerance.

Borg pre/post (target <5/10) **SpO_2 pre/post** **HR recovery at 1 minute**

Outcome Measure: 1MSTS repetition count — reassess at discharge and at 6-week outpatient follow-up to track recovery.

5 Treatment Plan

INITIAL PHASE — Weeks 1–2 (Acute Exacerbation)

AIRWAY CLEARANCE: ACTIVE CYCLE OF BREATHING TECHNIQUE (ACBT)

- **Rationale:** ACBT is the gold-standard ACT for bronchiectasis — evidence-based, adaptable, and requires no equipment, so Ms. Collins can use it independently at home long-term.
- **How to perform:** Three phases in a cycle, repeated 3–4 times per session: *(1) Breathing Control* — 3–4 relaxed diaphragmatic breaths at tidal volume, shoulders relaxed, to prevent bronchospasm. *(2) Thoracic Expansion Exercises (TEE)* — 3–4 slow, deep breaths with a 3-second inspiratory hold to open collateral ventilation channels and get air behind secretions. *(3) Forced Expiration Technique (FET/Huff)* — 1–2 medium-volume huffs (open glottis, "huhhh" sound — not a cough) with glottis open, followed by breathing control. Complete 2–3 sessions per day. Optimal position: upright sitting or gravity-assisted drainage position (e.g., 15° postural drainage if tolerated).

REGRESSION: Breathing control only if too fatigued or SpO_2 unstable. Shorten to 10-minute sessions. Assist manually with expiratory vibrations if patient cannot generate adequate expiratory flow.

PROGRESSION: Add gravity-assisted postural drainage positions. Introduce oscillatory PEP device (Flutter or Acapella) as an adjunct once secretions are partially mobilised. Increase to 20-minute sessions as fatigue improves.

BREATHING EXERCISE: DIAPHRAGMATIC BREATHING WITH LATERAL COSTAL EXPANSION

- **How to perform:** Patient in semi-reclined position. Place hands on lower lateral rib cage. Breathe in slowly through nose — feel ribs expand laterally, pushing hands outward. Breathe out through relaxed mouth or pursed lips. 5–8 breaths, 2–3 sets. Emphasis on preventing apical accessory muscle breathing and promoting diaphragmatic excursion.

REGRESSION: Reduce to 1–2 sets. Use as rest and recovery breathing between ACT phases only.

PROGRESSION: Add 3-second inspiratory hold. Progress to sitting upright in chair, then standing. Incorporate into functional activities (e.g., breathe during light walks).

EXERCISE / MOBILITY: EARLY GRADUATED MOBILISATION

- **Starting level:** Sitting out of bed in a chair for meals. Supervised ambulation in the room with rest breaks. Target Borg RPE 3–4/10. SpO_2 monitoring throughout. Encourage at least 2 short mobility sessions per day.

REGRESSION: Seated bed exercises — ankle pumps, heel slides, seated marching, shoulder rolls — if unable to tolerate standing.

PROGRESSION: Walking in the corridor (increasing distance) → supervised stair practice → community-level walking as discharge approaches.

PATIENT EDUCATION

- **ACBT technique and why the huff is NOT a cough:** Explain that huffing uses a mid-lung volume with an open glottis and is far more effective at moving peripheral secretions than coughing. Repeated harsh coughing causes fatigue, airway irritation, and potential airway collapse without improving clearance.
- **Hydration:** Adequate fluid intake (6–8 glasses/day unless contraindicated) reduces sputum viscosity and makes it easier to clear. Explain this directly connects to why ACT is currently less effective.
- **Exacerbation action plan — when to seek help:** Change in sputum colour to dark green/brown, haemoptysis (any blood), SpO_2 <92% at rest, or feeling significantly worse than usual baseline. Early treatment prevents hospitalisation.
- **Long-term importance of daily ACT:** Emphasise that ACT must continue every day even when feeling well — not just during exacerbations. Daily airway clearance breaks the vicious cycle of infection and slows progressive structural damage.

Progression Criteria (to move to rehabilitation phase): SpO_2 ≥94% on room air at rest · Borg ≤4/10 during supervised walking · RR ≤20 bpm at rest · Sputum volume trending back toward usual baseline · Patient independently performing correct ACBT technique

LONG-TERM PHASE — Weeks 4–12

- **Pulmonary Rehabilitation:** Refer to structured PR programme (minimum 8 weeks, twice weekly supervised sessions). Aerobic exercise training (walking, cycling) plus resistance training. Evidence shows PR significantly improves exercise capacity, QoL, and reduces exacerbation frequency in bronchiectasis.
- **Self-Management:** Reinforce daily independent ACBT (15–20 minutes/day). Patient maintains a written sputum diary. Review ACT technique and device use (e.g., Acapella) at each outpatient appointment.
- **Discharge Criteria:** Return to baseline sputum volume and character, SpO_2 ≥94% on room air, 1MSTS ≥ admission baseline, patient independently and correctly performing ACBT, and GP / respiratory outpatient follow-up within 2–4 weeks arranged.

⭐ Examiner Demonstration — Show These Techniques in the Room

- **ACBT — all three phases:** Demonstrate breathing control (relaxed tidal breathing), followed by 3 slow deep thoracic expansion exercises with a 3-second hold, then a medium-volume huff with open glottis (mouth open, "huhhh" — not a sharp cough). Show the cycle twice and explain each phase's purpose. Explicitly say: "I would now guide Ms. Collins through this and watch her perform it to correct technique."
- **Gravity-Assisted Drainage Positioning:** Demonstrate a 15° head-down / foot-up position (tipping bed or using a pillow wedge) for lower lobe drainage. Explain: "Gravity helps mucus move from the dilated peripheral airways toward the central airways where it can be cleared by huffing — this is especially important when sputum is viscous and hard to move."

Mr. John Harris

72-year-old Male · HFrEF — Decompensated · Secondary to Ischaemic Heart Disease · CXR: Pulmonary Oedema

CASE D

1 Subjective Assessment

CONSENT / OPENING STATEMENT

"Good morning Mr. Harris, my name is [Name] and I'm a physiotherapist. I've been asked to see you today to assess your breathing and help with your mobility and activity. I'll ask you some questions about how you've been feeling, and then I may do a gentle assessment. Nothing I do should be painful. Please

tell me if you feel breathless or unwell at any point, and we can stop. Do you have any questions? Are you happy for me to continue?"

KEY QUESTIONS + CLINICAL REASONING

- **How many pillows do you need to sleep, and has this changed over the past week?** *(Orthopnoea is a hallmark of left heart failure. Lying flat redistributes venous blood from the peripheries centrally, increasing pulmonary venous pressure and worsening oedema. Worsening orthopnoea signals progressive decompensation and helps me understand the severity of fluid overload.)*
- **Do you wake from sleep gasping for breath, usually an hour or two after lying down?** *(Paroxysmal nocturnal dyspnoea is a classic and concerning sign of acute pulmonary oedema — it indicates significant decompensation and is a red flag requiring medical attention before physiotherapy exercise.)*
- **How far can you walk on flat ground before you have to stop, and how does that compare to 2 weeks ago?** *(Quantifies functional deterioration — allows me to establish pre-admission baseline and classify NYHA functional class, which directly guides my exercise prescription intensity and realistic goals.)*
- **Have you been weighing yourself daily? Have you noticed a sudden increase in weight recently?** *(Rapid weight gain — 2 kg or more in 2–3 days — is the most sensitive early marker of fluid retention in heart failure. Understanding weight trajectory helps me gauge the degree of decompensation and the likely response to current diuretic therapy.)*
- **Have you had any chest pain, palpitations, or episodes of feeling faint?** *(Given his ischaemic aetiology, chest pain may indicate ongoing ischaemia or ACS — a contraindication to exercise. Palpitations suggest arrhythmia, which is common in HFrEF and may be haemodynamically significant. Both require urgent medical assessment before physiotherapy.)*
- **Have there been any recent changes to your medications, or have you had difficulty taking them consistently?** *(Non- adherence to diuretics is one of the most common precipitants of decompensation. Recent dose changes may explain the acute deterioration. Medication adherence also gives me context for the patient's self-management capacity and literacy.)*

RED FLAGS

Acute pulmonary oedema (sudden severe dyspnoea, SpO_2 <88–90%, pink frothy sputum)

Immediate life threat — fluid flooding the alveoli prevents effective gas exchange. Patient may deteriorate rapidly and require NIV or emergency intubation.

Action: Cease all physiotherapy immediately. Sit patient fully upright. Apply supplemental O_2. Call medical emergency. Do not leave patient alone.

Chest pain — new or changing (possible ACS)

Given his ischaemic heart disease aetiology, any new chest pain — particularly exertional — must be treated as potential ACS until ruled out. Exercise during active ischaemia can cause fatal arrhythmia.

Action: Stop all activity immediately. Obtain 12-lead ECG urgently. Alert medical team. Do not resume exercise until cleared.

Significant arrhythmia (new AF, rapid ventricular response, VT, HR >120 or <40 bpm at rest)

Arrhythmias are common in HFrEF and can cause haemodynamic compromise, especially with any physical exertion. Exercise is unsafe until rhythm is controlled.

Action: Defer all exercise. Check current ECG and vital signs. Notify medical team promptly.

Haemodynamic instability (SBP <90 mmHg, or drop >20 mmHg with position change)

Indicates severely reduced cardiac output — exercise will further reduce systemic perfusion and risk syncope, renal failure, and cardiac arrest.

Action: No exercise. Position for comfort (semi-reclined or sitting). Alert medical team. Reassess only when stable for >2 hours.

PROBLEM LIST

PHYSICAL IMPAIRMENTS

- Reduced cardiac output (impaired EF)
- Pulmonary oedema — impaired alveolar gas exchange
- Bilateral peripheral oedema — reduced ankle mobility
- Orthopnoea — positional dyspnoea
- Skeletal muscle deconditioning / cardiac cachexia
- Impaired peripheral oxygen delivery

FUNCTIONAL ACTIVITY LIMITATIONS

- Severely reduced exercise tolerance
- Dyspnoea completing ADLs (showering, dressing)
- Sleep disruption from orthopnoea / PND
- Reduced walking distance / community mobility
- Social isolation and reduced quality of life

2 Pathophysiology

In **HFrEF secondary to ischaemic heart disease**, prior myocardial infarction has caused irreversible loss of contractile myocardium — replaced by non-compliant fibrous scar tissue. This reduces the left ventricle's stroke volume and ejection fraction (typically <40%), meaning the heart pumps inadequate blood forward to meet the body's metabolic demands despite normal or elevated venous return.

As forward cardiac output falls, compensatory neurohormonal mechanisms activate — the sympathetic nervous system increases heart rate and peripheral vasoconstriction, and the renin-angiotensin-aldosterone system (RAAS) promotes sodium and water retention. While initially adaptive, **chronic RAAS activation causes progressive fluid overload**, increasing both preload and afterload on the already failing ventricle — perpetuating the cycle of deterioration.

The elevated left ventricular end-diastolic pressure is transmitted retrogradely through the left atrium to the pulmonary veins, raising **pulmonary capillary hydrostatic pressure**. When this exceeds the plasma oncotic pressure (approximately 25 mmHg), fluid leaks from pulmonary capillaries into the interstitium and — in severe decompensation — into the alveolar spaces. This **pulmonary oedema** impairs gas diffusion, causes V/Q mismatch, reduces lung compliance, and substantially increases the work of breathing.

Simultaneously, the same mechanism in the systemic venous circulation causes **bilateral peripheral oedema**. When Mr. Harris lies flat, this peripheral fluid is mobilised centrally, acutely increasing venous return and worsening pulmonary venous congestion — the physiological basis of orthopnoea and paroxysmal nocturnal dyspnoea.

Why Mr. Harris fits: His known HFrEF with ischaemic aetiology, 1-week history of worsening SOB, bilateral ankle oedema, orthopnoea, and CXR confirming pulmonary oedema are the textbook presentation of acute decompensated heart failure. His supplemental oxygen requirement reflects impaired alveolar gas exchange, and his fatigue reflects both reduced cardiac output to working muscles and skeletal muscle deconditioning from chronic low-output state.

3 Differential Diagnoses

Differential 1: COPD Exacerbation

Supporting features

- 72-year-old male — typical demographic for COPD
- Worsening dyspnoea and markedly reduced exercise tolerance
- CXR changes may superficially overlap (basal changes)
- Fatigue and reduced functional capacity common to both

Differential 2: Bilateral Pleural Effusions (non-cardiac cause)

Supporting features

- Bilateral CXR changes and progressive dyspnoea
- Orthopnoea possible with large bilateral effusions
- Bilateral ankle oedema if hypoalbuminaemia or renal disease
- Reduced exercise tolerance

I favour Decompensated HFrEF over COPD Exacerbation because Mr. Harris has a confirmed diagnosis of HFrEF with reduced EF secondary to ischaemic heart disease, and his presentation — orthopnoea, bilateral pitting ankle oedema, CXR showing pulmonary oedema (not hyperinflation, air trapping, or consolidation), and response to diuretic therapy — is mechanistically explained by cardiac failure. A COPD

exacerbation would typically feature productive cough, wheeze, and hyperinflation on CXR, none of which are identified as primary features here.

I favour Decompensated HFrEF over Isolated Pleural Effusions because the pattern on CXR (interstitial and alveolar pulmonary oedema) combined with orthopnoea, ankle oedema, and a known failing heart indicates the primary cardiac mechanism is driving both the pulmonary and systemic fluid accumulation. Pleural effusions in isolation would not explain the interstitial oedema pattern on CXR nor the full clinical picture of systemic venous hypertension.

4 Objective Assessment — Special Tests

Test 1: Resting Vital Signs + Orthostatic Blood Pressure Assessment

- **What it measures:** Haemodynamic stability — determines whether it is safe to proceed with any physiotherapy intervention. This is the non-negotiable first step in every decompensated HF session.
- **How to perform:** Record HR, BP, SpO_2, RR, and temperature at rest in supine, then sitting, then standing (if tolerated). Wait 2 minutes between positions. Note current supplemental O_2 flow rate and delivery device. Record Borg dyspnoea score. Auscultate heart and lungs. Inspect for peripheral oedema (grade 1–4+). Document baseline before any intervention.
- **Unsafe to exercise if:** SpO_2 <90% on supplemental O_2, SBP <90 mmHg, HR >120 or <40 bpm, RR >30 bpm, new chest pain, or symptomatic orthostatic hypotension (SBP drop >20 mmHg with dizziness).
- **Why I select this:** I choose vital signs assessment as my first test in every HF patient because the physiotherapy treatment plan — exercise intensity, positioning, duration — is entirely dependent on haemodynamic stability. In decompensated HFrEF, this changes session to session and I cannot prescribe exercise safely without confirming it first.

Expected Auscultation: Bibasal fine late-inspiratory crackles — these reflect fluid in the alveoli and interstitium. Critically, these do NOT clear with coughing (distinguishing them from bronchiectasis crackles). Diminished air entry at bases if pleural effusions are present. Possible third heart sound (S3 gallop) — pathognomonic of volume overload in HFrEF. Possible wheeze only if concurrent obstructive airway disease.

RR: 20–28 bpm (tachypnoeic) **SpO_2: 88–94% on O_2** **Accessory muscle use: moderate**

HR: assess for tachycardia/arrhythmia

Outcome Measure: Full vital signs documented at every session — track the trend across admission as a direct marker of medical response to diuresis.

Test 2: 1-Minute Sit-to-Stand Test (1MSTS)

- **What it measures:** Functional exercise capacity and physiological response to exertion — allows safe, brief quantification of cardiovascular and muscular functional reserve without the haemodynamic demands of a 6MWT.
- **How to perform:** Standard chair (no armrests, approximately 45 cm). Arms crossed on chest. On "Go," patient stands fully and returns to seated as many times as possible in 60 seconds. Count complete repetitions. Record HR, SpO_2, and Borg dyspnoea + fatigue scores immediately before and within 30 seconds of finishing. Have O_2 available. Stop criteria: SpO_2 <88%, chest pain, dizziness, SBP drop, Borg >6/10, or patient request.
- **Positive / Impaired finding:** <15 repetitions (normative ≈ 20–25 for 72M) = significantly reduced functional capacity. SpO_2 drop >4% = exercise-induced desaturation — supplemental O_2 required during all exercise sessions.
- **Why I select this:** I choose the 1MSTS because a full 6MWT is unsafe and inappropriate in the acute decompensated phase of HFrEF — it requires a corridor, is 6 minutes long, and cannot be stopped as easily if the patient deteriorates. The 1MSTS requires only a chair, takes 60 seconds, is equally valid as a baseline functional measure, and provides the data I need to prescribe exercise intensity safely.

Expected Auscultation (post-test): Possible exacerbation of bibasal crackles if exertion worsens cardiac preload transiently. Any new wheeze or dramatic increase in crackles is a sign of exercise intolerance — stop and reassess haemodynamics. New S3 post-exertion warrants urgent review.

Borg pre/post (target ≤5/10) **SpO_2 pre/post (target ≥92%)** **HR recovery at 1 minute**

Outcome Measure: 1MSTS repetition count — repeat at discharge and at 6-week cardiac rehab review. Minimum Clinically Important Difference ≈ 3–4 repetitions.

Test 3: NYHA Functional Class Classification

- **What it measures:** Standardised clinical classification of functional limitation severity in heart failure — universally recognised, enables communication with the cardiology team, and directly informs exercise prescription intensity.
- **How to perform:** Through structured subjective questioning — ask about symptoms during: (1) ordinary activity (Class I vs II), (2) less than ordinary activity (Class III), (3) at rest (Class IV). Classify: *Class I* = no symptoms with ordinary activity; *Class II* = slight limitation; *Class III* = marked limitation, comfortable only at rest; *Class IV* = symptoms at rest, any activity worsens discomfort.
- **Positive / Impaired finding:** Mr. Harris — worsening SOB over 1 week, bilateral oedema, orthopnoea, reduced exercise tolerance = likely NYHA Class III (marked limitation). Class IV if symptomatic at rest — exercise is contraindicated until medically stabilised.
- **Why I select this:** I use NYHA classification because it is the standardised language shared between physiotherapy and cardiology. It gives me a baseline that can be directly compared at discharge and follow-up, and it tells the team at a glance how much this patient's life is limited by their cardiac disease. NYHA Class also determines exercise prescription boundaries — Class III warrants very low intensity exercise with close monitoring.

Expected Auscultation: Bibasal fine crackles at rest — non-clearing. Reduced air entry at bilateral bases if effusions present. Possible S3 on cardiac auscultation. No wheeze expected unless concurrent lung disease.

RR at rest: 20–25 bpm | **SpO_2 on prescribed O_2** | **Ankle oedema grade (1–4+)**

Outcome Measure: NYHA Class at admission vs. discharge. Target: improvement of ≥1 class by discharge. Long-term target: NYHA Class I–II with optimal medical therapy and cardiac rehabilitation.

5 Treatment Plan

INITIAL PHASE — Weeks 1–2 (Acute Decompensation — Medical Stabilisation Priority)

POSITIONING — PRIMARY INTERVENTION (NOT AIRWAY CLEARANCE)

- **Important clinical note:** Pulmonary oedema is a cardiac/fluid problem — NOT a secretion-clearance problem. The physiotherapy role in the acute phase prioritises positioning, breathing strategies, and very gentle exercise, not ACT.
- **How to perform:** Position Mr. Harris in **high sitting (60–90°, legs dependent)**. Gravity reduces venous return to the right heart, lowering right-sided preload, which reduces pulmonary capillary hydrostatic pressure and the rate of fluid transudation into the alveoli. This is a direct, immediate intervention for dyspnoea relief. Legs should remain **dependent** (not elevated) in the acute phase — elevating legs increases venous return and can acutely worsen pulmonary oedema.

REGRESSION: Head of bed 30–45° if patient cannot maintain full sitting. Wedge pillow for positional support. Frequent position changes to prevent pressure injury.

PROGRESSION: Sitting out of bed in a high-backed chair (90°) → sitting with legs down for increasing periods → supervised standing at bedside.

BREATHING EXERCISE: PURSED-LIP BREATHING (PLB)

- **How to perform:** Patient in high sitting with supported arms (resting on pillows or chair armrests to offload accessory muscles). Inhale slowly through nose for 2 counts. Exhale through pursed lips (as if blowing through a straw) for 4 counts — 2:1 ratio. This generates a mild auto-PEEP effect, which splints small airways open, reduces air trapping, slows respiratory rate, and improves alveolar ventilation. Teach 5–10 breaths, 2–3 sets, 2–3 times daily, or on demand during exertional dyspnoea.

REGRESSION: Emphasise slow controlled nasal breathing with focus on relaxed exhalation only. Reduce to 3–5 breaths if fatiguing. Prioritise rest between sets.

PROGRESSION: Practise during sit-to-stand transfers and short walks — "breathe out on the effort." Use as a dyspnoea self-management tool during daily activities.

EXERCISE / MOBILITY: VERY LOW-INTENSITY GRADUATED MOBILISATION

- **When to start:** Only when haemodynamically stable for >24–48 hours (SBP >90 mmHg, HR 50–100 bpm, SpO_2 ≥90% on supplemental O_2, no chest pain).
- **Starting level:** Bed exercises — ankle pumps, heel slides, seated marching, shoulder rolls — progressing to supervised sit-to-stand (2–3 repetitions) then standing at bedside. Target Borg RPE 2–3/10. SpO_2 and HR monitoring throughout. Maximum 10–15 minutes initially.

REGRESSION: Passive limb movements and positioning only if too unwell for active exercise. Bed-based deep breathing to maintain respiratory muscle function.

PROGRESSION: Supervised corridor walking 10–20 m → increasing daily distance → stair assessment before discharge → community walking programme.

Stop exercise immediately if: *SpO_2 <88%, SBP <90 mmHg or drops >20 mmHg, new chest pain or palpitations, Borg >6/10, dizziness or pre-syncope, or new arrhythmia on telemetry.*

PATIENT EDUCATION

- **Daily weight monitoring:** Weigh each morning after toileting, before breakfast. Record in a diary. If weight increases >2 kg in 2 days, contact GP or heart failure nurse — this is an early warning sign of fluid retention before symptoms become severe. This single habit prevents many hospitalisations.
- **Fluid and sodium management:** Adhere to any prescribed fluid restriction (commonly 1.5–2 L/day). Reduce dietary salt — excess sodium causes fluid retention, which worsens oedema and breathlessness directly.
- **Medication adherence:** Never skip diuretics, ACE inhibitors, or beta-blockers even when feeling better. These medications protect the heart muscle and reduce long-term mortality — not just symptom relief.
- **Energy conservation and pacing:** Plan rest breaks before and after activities. Use pursed-lip breathing during exertion. Avoid rushing. Recognise warning signs of decompensation: increasing ankle swelling, needing more pillows, waking breathless at night. Seek help early.

Progression Criteria (to move to rehabilitation phase): SpO_2 ≥92% on ≤2 L/min O_2 (or room air) · Borg ≤4/10 during supervised walking · RR ≤22 bpm at rest · No chest pain or symptomatic hypotension with exertion · Haemodynamically stable >48 hours · NYHA Class improved ≥1 from admission · Patient demonstrates daily weight monitoring skill

LONG-TERM PHASE — Weeks 4–12

- **Cardiac Rehabilitation:** Refer to structured cardiac rehabilitation programme (ESC/ACPRC guidelines) — minimum 8–12 weeks, supervised aerobic exercise 3–5 days/week. Begin with 10–15 minutes continuous moderate-intensity exercise (Borg 3–5/10), progressing to 30

minutes. Strong evidence for reduced re-hospitalisation, improved exercise capacity, and mortality benefit in HFrEF.

- **Resistance Training:** Low-to-moderate resistance exercises targeting upper and lower limbs (resistance bands, light weights). Start at 30–40% 1RM, 2 sets × 10–15 repetitions. Addresses skeletal muscle deconditioning — a major contributor to exercise intolerance in HFrEF beyond cardiac output limitations.
- **Self-Management:** Patient maintains a daily weight and symptoms diary. Written heart failure action plan (when to call nurse, when to go to ED) shared with GP, cardiologist, heart failure nurse, and patient.
- **Discharge Criteria:** Haemodynamically stable on oral medications, SpO_2 ≥92% on room air, independently mobile to functional baseline, patient demonstrates self-monitoring skills, and heart failure nurse / cardiology follow-up within 2 weeks arranged before leaving hospital.

⭑ Examiner Demonstration — Show These Techniques in the Room

- **Pursed-Lip Breathing:** Demonstrate from the patient's position — inhale slowly through nose (2 counts), exhale through gently pursed lips as if blowing through a straw (4 counts). Show relaxed shoulders, no breath-holding, no forceful exhalation. Say: "I would coach Mr. Harris through 5–6 breaths and teach him to use this independently when walking or transferring — breathe out on the hard part." Demonstrate the 2:1 ratio clearly.
- **High-Sitting Positioning for Dyspnoea Relief:** Demonstrate sitting upright at 90° in bed with legs dependent, supported by pillows. Explain the rationale: "Gravity reduces venous return from the peripheries, lowering preload on the right heart, which reduces pulmonary venous pressure and slows fluid accumulation in the alveoli. This is our first-line physiotherapy intervention for acute dyspnoea in decompensated heart failure — and it costs nothing."